Lunar Descent Plant and Flight Software: Mathematics

1 Conventions

SI units throughout: meters, seconds, kilograms, radians. Frame: origin at the Moon's center, z along the Moon's spin axis, axes fixed relative to the stars (Moon-centered inertial). The Moon does not move. Source: `workloads/moon/common/`.

2 Vectors

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ and scalar s:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{1}$$

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x) \tag{2}$$

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} \tag{3}$$

The cross product follows the right-hand rule: $\hat{x} \times \hat{y} = \hat{z}$. Code: `vec3.hpp`. Tests: `test_vec3.cpp`.

3 Constants

Symbol	Value	Unit	Source
μ_M	$4.902800066 \times 10^{12}$	$\mathrm{m}^3/\mathrm{s}^2$	JPL DE430
R_M	1 737 400	m	mean radius
ω_M	2.6616995×10^{-6}	rad/s	sidereal spin, one turn per 27.321661 days
g_0	9.80665	m/s^2	standard gravity

The spin vector is $\boldsymbol{\omega}_M = (0, 0, \omega_M)$. The ground under a point $\mathbf{r}$ moves at $\boldsymbol{\omega}_M \times \mathbf{r}$, which is $\omega_M R_M = 4.624$ m/s at the equator. Code: `constants.hpp`.

4 Gravity

With $r = \|\mathbf{r}\|$:

$$\mathbf{g} = -\mu_M \frac{\mathbf{r}}{r^3} \tag{4}$$

At the surface, $\mu_M / R_M^2 = 1.6242$ m/s^2. Earth's tidal acceleration near the Moon is on the order of 10^{-5} m/s^2 (estimated) and is not modeled. Code: `dynamics.cpp`. Tests: `test_dynamics.cpp`.

5 Engine and actuator limits

The flight software commands a throttle τ and a direction $\mathbf{d}$. The plant applies the limits of a real engine:

$$\tau' = \begin{cases} 1 & \tau > 1 \\ \tau & 0 < \tau \leq 1 \\ 0 & \text{otherwise (including NaN)} \end{cases} \tag{5}$$

The engine is off ($\tau' = 0$) when $\|\mathbf{d}\|$ is zero, infinite, or NaN, or when the mass has reached the dry mass m_{dry}. Otherwise:

$$F = \tau' F_{\max} \tag{6}$$

$$\mathbf{a}_T = \frac{F}{m} \frac{\mathbf{d}}{\|\mathbf{d}\|} \tag{7}$$

$$\dot{m} = -\frac{F}{I_{sp}\, g_0} \tag{8}$$

The direction is normalized, so a corrupted command can point the engine anywhere but cannot push harder than full thrust. These limits are physics of the plant, not a flight software defense.

6 Equations of motion

$$\dot{\mathbf{r}} = \mathbf{v} \tag{9}$$

$$\dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}) + \mathbf{a}_T \tag{10}$$

$$\dot{m} = -\frac{F}{I_{sp}\, g_0} \tag{11}$$

The system does not depend on time explicitly. Code: `dynamics.cpp`, function `derivative`.

7 Numerical integration (RK4)

Write the state as $\mathbf{y} = (\mathbf{r}, \mathbf{v}, m)$ and the equations of motion as $\dot{\mathbf{y}} = f(\mathbf{y})$. The classical fourth-order Runge–Kutta step of size h:

$$\mathbf{k}_1 = f(\mathbf{y}_n) \tag{12}$$

$$\mathbf{k}_2 = f(\mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1) \tag{13}$$

$$\mathbf{k}_3 = f(\mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_2) \tag{14}$$

$$\mathbf{k}_4 = f(\mathbf{y}_n + h\,\mathbf{k}_3) \tag{15}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{h}{6}\,(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \tag{16}$$

The engine command is held constant for the whole step, as flight software sets one command per tick. If $m_{n+1} < m_{\text{dry}}$, it is set to m_{dry}; the step still applied thrust for its full length, so the error is at most one step of thrust.

Validation. With no thrust, specific orbital energy $\mathcal{E} = \frac{1}{2}\|\mathbf{v}\|^2 - \mu_M/r$ and angular momentum $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ are conserved. Over one 100 km circular lunar orbit (period 7067.5 s) in 5000 steps, measured drift is 2.1×10^{-15} in $\mathcal{E}$ and 8.9×10^{-16} in $\|\mathbf{h}\|$, at the level of double rounding, and the spacecraft returns to within a millimeter of its start. Forward Euler on the same orbit drifts 1.5×10^{-2} in $\mathcal{E}$ and returns 138 km off. Code: `integrator.cpp`. Tests: `test_integrator.cpp`.

8 Navigation (flight software)

The flight software receives the Moon-centered state $(\mathbf{r}, \mathbf{v})$ and derives the ground-relative state:

$$\mathbf{u} = \mathbf{v} - \boldsymbol{\omega}_M \times \mathbf{r} \qquad \text{velocity relative to the ground} \tag{17}$$

$$\hat{\mathbf{n}} = \mathbf{r}/\|\mathbf{r}\| \qquad \text{local up} \tag{18}$$

$$h = \|\mathbf{r}\| - R_M \qquad \text{altitude} \tag{19}$$

$$v_\uparrow = \mathbf{u} \cdot \hat{\mathbf{n}} \qquad \text{vertical speed, positive up} \tag{20}$$

$$v_\leftrightarrow = \|\mathbf{u} - v_\uparrow \hat{\mathbf{n}}\| \qquad \text{horizontal speed} \tag{21}$$

This stage runs before guidance, so a fault here reaches every TMR replica.

Limitations. The real lunar spin axis is tilted about 1.5° from the orbit normal (Cassini's laws); the model fixes it along z. Altitude is above a sphere of radius R_M; there is no terrain. Code: `fsw/navigation.cpp`. Tests: `test_navigation.cpp`.

9 Guidance (flight software)

Inputs: the navigation state and the mass m. Output: throttle τ and direction $\mathbf{d}$; the plant applies the actuator limits.

Braking. Full throttle, $\tau = 1$, split between altitude control and horizontal deceleration. With $a_{\max} = F_{\max}/m$, $w = v_\leftrightarrow$, $r = R_M + h$, and $h_b = 1500$ m:

$$t_{\text{stop}} = w/a_{\max} \tag{22}$$

$$v_b^* = -\frac{h - h_b}{\max(t_{\text{stop}},\ 1\text{ s})} \tag{23}$$

$$a_\uparrow = \text{clamp}\left(\frac{\mu_M}{r^2} - \frac{w^2}{r} + k_v\,(v_b^* - v_\uparrow),\ 0,\ a_{\max}\right) \tag{24}$$

$$a_\leftrightarrow = \sqrt{(a_{\max} - a_\uparrow)(a_{\max} + a_\uparrow)} \tag{25}$$

$$\mathbf{d} = a_\uparrow \hat{\mathbf{n}} - \frac{a_\leftrightarrow}{w}\,(\mathbf{u} - v_\uparrow \hat{\mathbf{n}}) \tag{26}$$

The sink-rate target brings the lander to h_b as its horizontal speed reaches zero. w^2/r is the support still provided by orbital motion. The factored form of $a_\leftrightarrow$ cannot go negative by rounding, because the clamp guarantees $a_\uparrow \le a_{\max}$. Pure retrograde braking from 15 km (no altitude control) was measured to strike the surface at 132 m/s with 726 m/s of horizontal speed remaining.

Switch. Braking ends, permanently, when $v_\leftrightarrow < 50$ m/s or $h < 500$ m, whichever comes first.

Final descent. Target vertical speed and wanted acceleration:

$$v^* = -\operatorname{clamp}\left(\frac{h}{10\ \mathrm{s}},\ 1,\ 30\right)\ \mathrm{m/s} \tag{27}$$

$$g = \frac{\mu_M}{(R_M + h)^2} \tag{28}$$

$$\mathbf{a}^* = (g + k_v\,(v^* - v_\uparrow))\,\hat{\mathbf{n}} - k_h\,(\mathbf{u} - v_\uparrow \hat{\mathbf{n}}) \tag{29}$$

with $k_v = 1.0\ \mathrm{s}^{-1}$ and $k_h = 0.5\ \mathrm{s}^{-1}$. If $\mathbf{a}^* \cdot \hat{\mathbf{n}} \le 0$ the engine is off (a lander cannot thrust toward the ground). Otherwise

$$\tau = \frac{\|\mathbf{a}^*\|\, m}{F_{\max}}, \qquad \mathbf{d} = \mathbf{a}^* \tag{30}$$

and the plant saturates τ at 1. On the target descent rate with no drift, $\tau = g\,m/F_{\max}$, the hover throttle.

Clean landing. Measured: touchdown at 527.8 s at 1.000 m/s descent and 0.000 m/s horizontal speed, with 1157.6 kg of propellant left; score 0.328. Two clean runs produce byte-identical traces. Code: `fsw/guidance.cpp`. Tests: `test_guidance.cpp`.

10 Simulation and landing judge

Start. Circular equatorial prograde orbit at $h_0 = 15\,\mathrm{km}$: $\mathbf{r}_0 = (R_M + h_0, 0, 0)$, $\mathbf{v}_0 = (0, \sqrt{\mu_M/(R_M + h_0)}, 0)$, $m_0 = 15\,200$ kg. The flight software and the plant step together at $\Delta t = 0.1$ s, for at most 3600 s.

Independence. The judge computes altitude and ground-relative velocity with the same formulas as navigation, in uninstrumented plant code. If it reused the flight software's navigation, a fault there would corrupt the judge as well.

Touchdown. When altitude changes sign within a step, from $h_a > 0$ to $h_b \le 0$, the crossing fraction is $f = h_a/(h_a - h_b)$, and touchdown speeds, time, and position are interpolated linearly at f.

Ground-fixed spot. With $\alpha = -\omega_M t$, the touchdown point is rotated back to the ground's orientation at $t = 0$:

$$\mathbf{s} = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{r} \tag{31}$$

so two runs landing on the same ground spot at different times give the same $\mathbf{s}$.

Score. With descent speed v, horizontal speed w, and distance d from the clean run's spot (zero for the clean run itself):

$$v_{\lim}(w) = 3.05 - (3.05 - 2.13)\frac{w}{1.22}, \qquad S = \max\left(\frac{v}{v_{\lim}(w)},\ \frac{w}{1.22},\ \frac{d}{1000}\right) \tag{32}$$

in m/s and m (ADR-0020). $S \le 1$ passes. No touchdown within the time limit gives $S = \infty$. The plant reports S as the run's final result. Code: `plant/judge.cpp`, `sim/main.cpp`. Tests: `test_judge.cpp`.